

9.3.12.

a. How many ways can the letters of the word *THEORY* be arranged in a row?

b. How many ways can the letters of the word *THEORY* be arranged in a row if T and H must remain next to each other as either TH or HT?

a. How many ways can the letters of the word *THEORY* be arranged in a row?

T, H, E, O, R, Y

There're 6-permutations of a set of six elements.

$$P(6, 6) = \frac{6!}{(6-6)!} = \frac{6!}{0!} = \frac{6!}{1} = 6!$$

Solution: a. There're $6!$ ways the letters of the word *THEORY* can be arranged in a row.

b. How many ways can the letters of the word *THEORY* be arranged in a row if T and H must remain next to each other as either TH or HT?

TH, E, O, R, Y

HT, E, O, R, Y

There're 5-permutations of a set of five elements.

$$P(5, 5) = \frac{5!}{(5-5)!} = \frac{5!}{0!} = \frac{5!}{1} = 5!$$

But we must also take into account that there're two different ways to arrange them. Therefore:

$$2 \times P(5, 5) = 2 \times 5!$$

Solution: b. There're $2 \times 5!$ ways to arrange the letters of the word *THEORY* in a row if T and H must remain next to each other as either TH or HT.